



Università
Ca' Foscari
Venezia

[CM0471] STATISTICAL INFERENCE AND LEARNING

ACADEMIC YEAR: 2025-26

**PROJECT: A DATA-DRIVEN STUDY OF AIR QUALITY IN NEW YORK CITIES USING
WEATHER AND POLLUTANTS**

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1. Objective

The dataset is a daily table at the city level that includes meteorological conditions, pollutant-specific AQI sub-indices, and air pollution readings. The goal is

- to model/predict daily AQI using weather + pollutants and
- to compare patterns between 2020 (COVID period) and 2024 (more normal period) because each row represents a single city on a single calendar day.

2. Dataset

2.1 Data Sources and Merging Datasets

- Pollutant data (EPA):** We downloaded daily CO, NO₂, SO₂, PM_{2.5}, O₃ data for New York State from [epa.gov](https://www.epa.gov), where each city has several monitoring stations. **City-day aggregation:** For each pollutant, we grouped by (date, city), took the mean concentration for different stations (typical city level) and the maximum daily AQI (worst air in that city that day). **Combine pollutants:** We merged all pollutant tables on (date, city) with an outer join, so any day where at least one pollutant is available is kept in the dataset.
- Weather Data:** We downloaded daily weather data for the cities Bronx, Erie, Monroe, and Queens from [Visual Crossing](https://www.visualcrossing.com) and we kept only the temperature, humidity and wind_speed column, and merged them with the pollutant data on (date, city) using an inner join (keeping only days with both pollution and weather).
- Target label:** For each city-day, We took the maximum AQI across pollutants and used it to label each day: (i) **healthy** if $\max_aqi \leq 50$, (ii) **unhealthy** if $\max_aqi \geq 51$

2.2 Dataset Description

The dataset covers two leap years (2020 and 2024) and four New York State cities: Bronx, Erie, Monroe, and Queens. Since each leap year has 366 days, the dataset contains: $4 \text{ cities} \times 366 \text{ days} =$

1464 rows per year, and $1464 (2020) + 1464 (2024) = 2928$ total rows. The columns are organized as follows:

1. **Date and location:** date (calendar date), city (Bronx/Erie/Monroe/Queens), and year (extracted from date: 2020 or 2024).
2. **Pollutant concentrations and AQI (daily mean values):** `co_mean(ppm)/co_aqi`, `no2_mean(ppb)/no2_aqi`, `so2_mean(ppb)/so2_aqi`, `pm2_5_mean(ug/m3)/pm2_5_aqi`, and `o3_mean(ppm)/o3_aqi`.
3. **Meteorological predictors (daily values):** temp, humidity, and windspeed.
4. **Derived target and labels:** `max_aqi` (regression target; the **maximum** pollutant AQI sub-index for that day, i.e., the “worst” pollutant determines the day’s AQI), `Main_Pollutant` (the pollutant achieving `max_aqi`), `aqi_level` (categorical label from `max_aqi`), and `aqi_level_bin` (numeric encoding: 0 = healthy, 1 = unhealthy).
5. **Missing values:** Weather variables and `max_aqi` are complete, while some pollutant columns have missing values: `co_mean/co_aqi` 10 ($\approx 0.3\%$), `no2_mean/no2_aqi` 86 ($\approx 2.9\%$), `so2_mean/so2_aqi` 30 ($\approx 1.0\%$), and `o3_mean/o3_aqi` 115 ($\approx 3.9\%$).

2.3 Data Visualization and Comparison of the Air Quality Pattern between 2020 and 2024

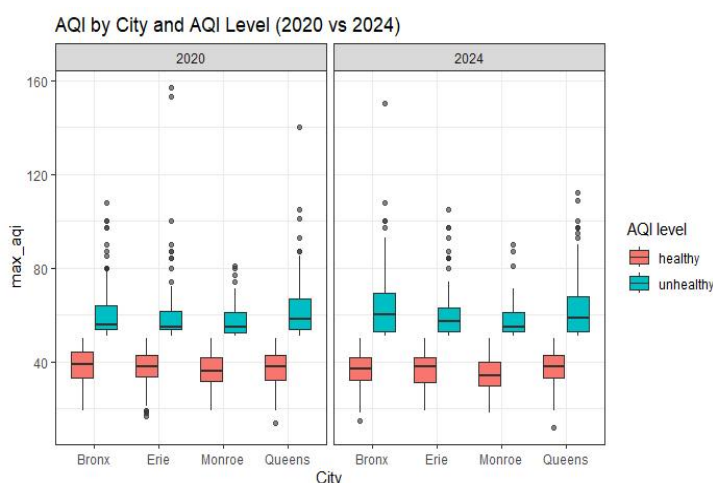


Figure 1: AQI by City and AQI Level (2020 vs 2024)

Figure 1 compares `max_aqi` in the Bronx, Erie, Monroe, and Queens (2020 vs 2024), categorizing it as healthy or unhealthy. In both years, the classes are clearly separated: healthy days are about ~ 35 - 40 , whereas unhealthy days are higher at ~ 55 - 65 . The healthy ranges appear to be quite comparable across cities from 2020 to 2024,

implying that baseline conditions did not change significantly; The key difference is that bad days are larger/more variable in 2024, particularly in the Bronx and Queens. Overall, Queens has the highest and broadest AQI distribution, whereas Monroe has the lowest and most consistent. Occasional significant spikes (~150-160) occurred in both years, indicating infrequent severe pollution episodes occur.

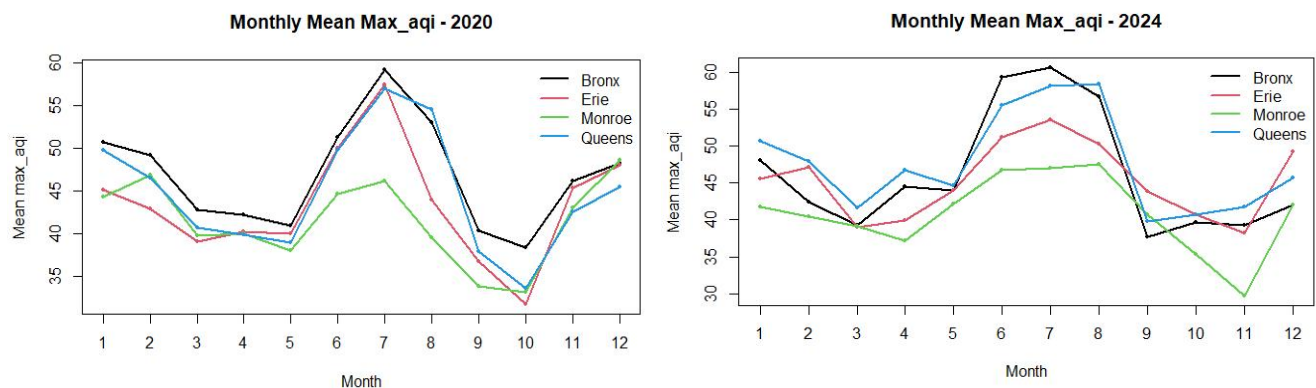


Figure 2: Monthly Mean Max Aqi (2020 vs 2024)

In figure 2, both 2020 and 2024 exhibit a distinct seasonal pattern in monthly mean max_aqi across all cities, with values peaking in the summer (Jun-Aug) and dropping to a minimum around September-October. In 2020, the peak is strongest in July, reaching roughly 58-60 in Bronx and ~55-58 in Queens/Erie. Monroe remains lower, peaking in the mid-40s. In 2024, the peak occurs earlier and wider (Jun-Aug): Bronx ~60 in Jun-Jul and Queens ~57-58 in Jul-Aug, whereas Erie peaks lower than 2020 (low-50s). After summer, both years decline, although Monroe dips more strongly in 2024, hitting ~30 in November, resulting in a similar summer maximum but smoother peak and more late-year variability. Overall, in 2024 the AQI is higher and wider as compared to 2020.

This bar chart in figure 3, compares number of healthy and unhealthy days per city in 2020 vs 2024. In 2020, the Bronx is the worst, with 137 unhealthy and 229 healthy, while Erie and Monroe are the

best (87 unhealthy and 279 healthy each), and Queens is in the middle (107 unhealthy, 259 good). In 2024, the pattern changes: Erie rises to 131 unhealthy (235 healthy), while Queens rises to 142

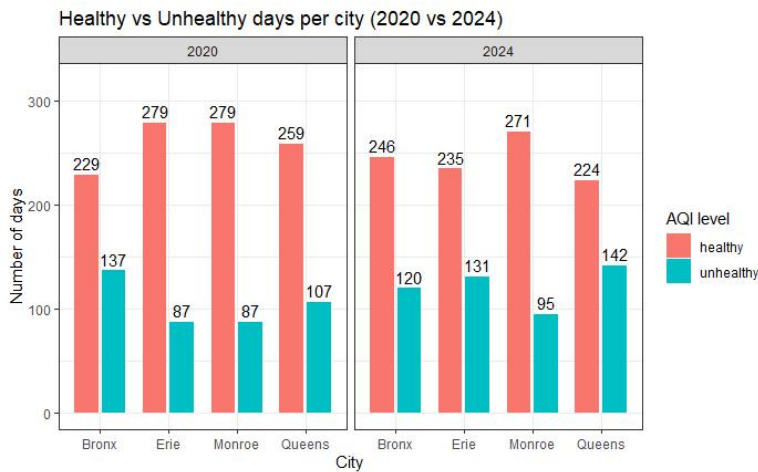


Figure 3: Healthy and Unhealthy days per city (2020 vs 2024)

unhealthy (224 healthy). Meanwhile, the Bronx improves (137 to 120 unhealthy, 229 to 246 healthy), whereas Monroe changes only marginally (279 to 271 healthy, 87 to 95 unhealthy). Overall, bad-air days are decreasing in the Bronx but increasing significantly in Queens and

Erie, indicating that the burden of unhealthy days has shifted among communities. Therefore, in 2024 the number of unhealthy days increased.

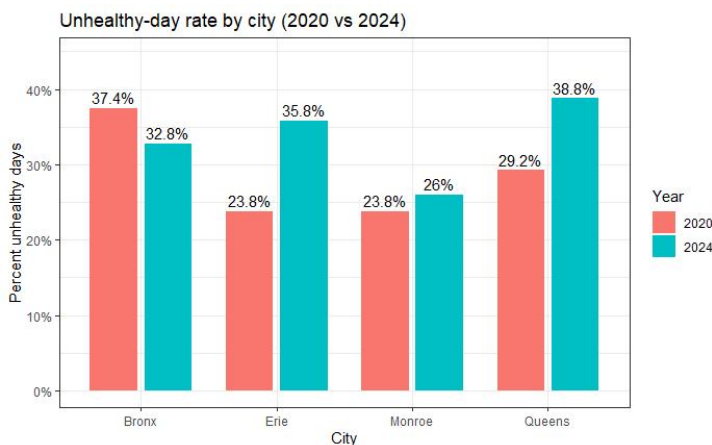


Figure 4: Unhealthy Day Rate by City (2020 vs 2024)

Figure 4 shows the percentages of the unhealthy days. Unhealthy-day rate fluctuates between 2020 and 2024 and varies by city. While the increase of unhealthy days in Erie dramatic from 23.8% to 35.8% (almost 12%), reaching a level comparable to the Bronx, the Bronx

improves, falling from 37.4% to 32.8%. Monroe has only slightly increased (23.8% → 26.0%) and is still the least impacted. In 2024, Queens experiences the worst growth, going from 29.2% to 38.8%. With the exception of the Bronx, all cities see an increase in the percentage of unhealthy days, which shifts the burden from the Bronx to Queens (and Erie). So, it can be concluded that the rate of

unhealthy days has been dramatically increased in 2024 in most of the cities and thus 2024 can be said a unhealthy year as compared to 2020.

2.4 Feature Engineering and Missing Values

To prevent leakage and maintain the time structure, the dataset was divided into two temporal segments: train (2020) and test (2024). A 12-level month factor ("01"–"12") was developed to reflect seasonality once the date column was changed to a proper date. The majority of missingness was found in pollutant variables, and a missingness was examined rather than eliminated. For NO₂/O₃/SO₂, GLM revealed systematic missingness (connected to city, season, and weather), whereas CO missingness seemed to be mostly random. We employed PCA imputation fitted on the 2020 training set and applied consistently to both train and test because dropping rows under patterned missingness can skew results and reduce sample size. This method takes advantage of the correlation structure between pollutants and meteorological variables rather than filling with a constant using mean, median or mode.

3 Model Selection

We started with a baseline linear regression (LM), but LM assumes each predictor affects AQI with a constant slope. In practice, key drivers such as PM_{2.5}, O₃, and temperature often show non-linear effects. To capture these realistic shapes while keeping interpretation clear, I switched to a Generalized Additive Model (GAM) using `mgcv`.

3.1 Baseline Linear Regression (LM)

Using the 2020 training set, we fit $\text{max_aqi} \sim \text{city} + \text{temp} + \text{humidity} + \text{windspeed} + \text{co_mean} + \text{no2_mean} + \text{so2_mean} + \text{pm2_5_mean} + \text{o3_mean}$. The fit was moderate ($R^2 = 0.7957$, adj. $R^2 = 0.7942$; residual SE = 6.40), but diagnostics showed clear assumption violations: the residuals-vs-fitted trend is roughly flat only up to fitted values around 60 and then drifts negative (missed nonlinearity and overprediction at high AQI); the Scale-Location plot fans out (heteroscedasticity); and the Q-Q Plot reveals a right-skewed upper tail with outliers (e.g., indices 1302, 1310 and 161).

ResidualPlots also indicates remaining nonlinearity, especially for o3_mean (test stat = 25.636, $p < 2.2e-16$) and for temp/co_mean/pm2_5_mean. Additionally, we also built different versions of LM, such as discarding influential points or transforming target using Box-Cox transformation; however no LM fit the dataset well. Therefore, LM was retained only as a baseline and we proceeded with GAMs including smooth terms and interactions. A GAM extends LM by replacing linear terms with smooth functions:

$$AQI = \beta_0 + \sum_j f_j(x_j) + (\text{parametric terms}) + \varepsilon$$

where each $f(\cdot)$ is a smooth curve learned from data (penalization prevents overfitting). This remains interpretable because each smooth plot shows how AQI changes with one predictor while others are fixed.

Step 1: Baseline GAM (nonlinear main effects): We fit an additive GAM on the original AQI scale, with city as a parametric factor (baseline city differences). Major nonlinear drivers were modeled with $s()$: $s(\text{PM2.5})$, $s(\text{O3})$, $s(\text{SO2})$, $s(\text{temp})$, while NO_2 , CO , humidity, windspeed were kept linear initially for simplicity. This already improved interpretability vs LM by revealing nonlinear exposure - response curves, especially for $\text{PM}_{2.5}$ and O_3 .

Step 2: Controlling smoothness with k (basis dimension): In GAM, k sets the maximum complexity of each smooth. Larger k allows more wiggles, but mgcv still penalizes so it does not automatically overfit. We began with $k=6$, then increased to $k=8$ and 10 for stronger nonlinear predictors (especially $\text{PM}_{2.5}$ and O_3 , and also other predictors when needed) because small k can underfit. When edf is close to k' , the model is using most of the allowed flexibility, so increasing k can be reasonable. This ensured smooths were flexible enough without forcing linearity.

Step 3: Box-Cox target transformation (variance/tails): AQI is often right-skewed with occasional extreme high-AQI days, causing (1) heavy-tail non-normal residuals (QQ-plot issues) and (2) heteroscedasticity (variance increasing with fitted values). We therefore applied a Box-Cox transformation and fit the GAM on the transformed response:

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \log(y) & \lambda = 0 \end{cases}$$

which often makes residuals more normal, stabilizes variance, and improves inference (SEs/tests) in a Gaussian GAM. Predictions were converted back to AQI using the inverse Box-Cox for evaluation.

Step 4: Adding seasonality (month effect): Because AQI is seasonal, We added the month term using a cyclic smooth $s(\text{month}, \text{bs}="cc")$ with the Box-Cox transformed target to ensure December connects smoothly to January. Adding month improved interpretability and reduced time-of-year residual structure.

Step 5: Adding interactions (tensor product smooths): Additive GAMs assume independent contributions, but environmental effects can interact: PM_{2.5} depends on windspeed (dispersion), O₃ and PM_{2.5} can jointly affect AQI, and the temp effect varies across months. We added tensor interactions using $ti()$: $ti(\text{pm2_5_mean}, \text{windspeed}, \dots)$, $ti(\text{o3_mean}, \text{pm2_5_mean}, \dots)$, and $ti(\text{month}, \text{temp}, \dots)$. Tensor products are suitable when predictors have different units/scales or when one is cyclic (month) and one is continuous (temp).

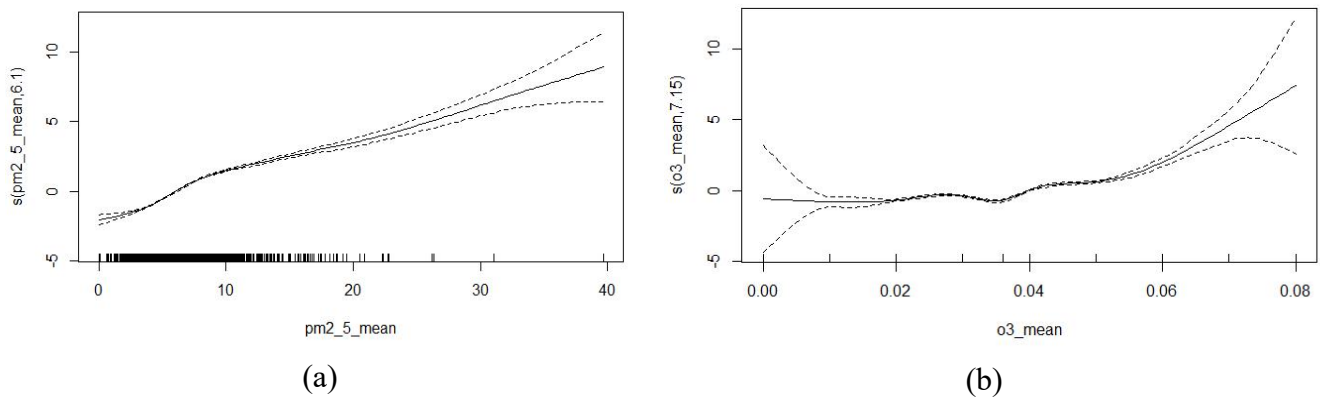
3.2 Final Model (GAM)

The final model is a Box-Cox GAM with (1) parametric city effects, (2) smooth main effects for major nonlinear drivers (PM_{2.5}, O₃, temp, etc.), and (3) key interaction surfaces via $ti()$ (pollutant-pollutant and month-temperature). It was selected for the best balance of interpretability (main-effect smooths + interaction surfaces), statistical validity (better residual behavior after transformation), and generalization (lower test RMSE vs simpler GAM variants).

4 Result and Discussion

This final Gaussian GAM (identity link) models the Box-Cox transformed AQI (y_{bc}) using city fixed effects, smooth nonlinear terms for key pollutants/weather, and two interaction surfaces. The fit is very strong (adj. $R^2 = 0.929$, 93.2% deviance explained, $n=1464$) and it also generalizes well to 2024 (RMSE = 4.2776 after back-transforming). In the parametric part, the intercept is highly

significant (9.93, $p < 2e-16$). Relative to the reference city, Erie (−0.365), Monroe (−0.226), and Queens (−0.160) all have significantly lower baseline AQI (all $p < 0.001$). Among linear predictors, NO₂ is significant ($\beta = 0.00467$, $p = 0.011$), while CO, humidity, and windspeed are not significant once smooths/interactions are included. For smooth terms, PM_{2.5} and O₃ are the dominant nonlinear drivers (edf≈6.10, $F=533$; edf≈7.15, $F=110.76$; both $p < 2e-16$). SO₂ adds little (edf=1, $p=0.422$). Temperature behaves essentially linear (edf=1) but remains significant ($p=0.003$). Crucially, both interactions are strongly supported: ti(O₃, PM_{2.5}) is highly significant (edf≈38.99, $F=25.61$, $p < 2e-16$), indicating pollutant synergy, and ti(month_num, temp) is significant (edf≈17.47, $p < 2e-16$), showing the temperature-AQI relationship varies seasonally (month treated cyclic).



In figure 5, the smooth $s(\text{pm2_5_mean}, \text{edf} \approx 6.1)$ for PM_{2.5} grows nearly monotonically (particularly after ~10), indicating that greater PM_{2.5} is associated with higher expected AQI, with broader uncertainty after ~30 when data are sparse. O₃ has a substantial non-linear "high ozone → much higher AQI" effect, with $s(\text{o3_mean}, \text{edf} \approx 7.15)$ remaining almost flat at low-mid values but rising abruptly around 0.05-0.08. The dashed bands are ± 2 SE, widening at the extremes because there are fewer observations there. $s(\text{so2_mean}, \text{edf} \approx 1)$ for SO₂ is essentially a flat line around zero, indicating that after additional variables are included in the model, SO₂ adds very little. After correcting for pollutants, $s(\text{temp}, \text{edf} \approx 1)$ for temperature is likewise nearly flat, suggesting no discernible additional influence. Lastly, a complex non-additive surface is displayed by the interaction $\text{ti}(\text{o3_mean}, \text{pm2_5_mean}, \text{edf} \approx 39)$. The influence of O₃ relies on PM_{2.5} level (and vice

versa), with outlines indicating places where their combined impact is much greater or lesser than a simple additive model. The GAM interaction $ti(\text{month_num}, \text{temp})$ ($\text{edf} \approx 17.5$) indicates that temperature has a seasonal influence. The contour shapes show non-linear month-temp regions, with significant impacts.

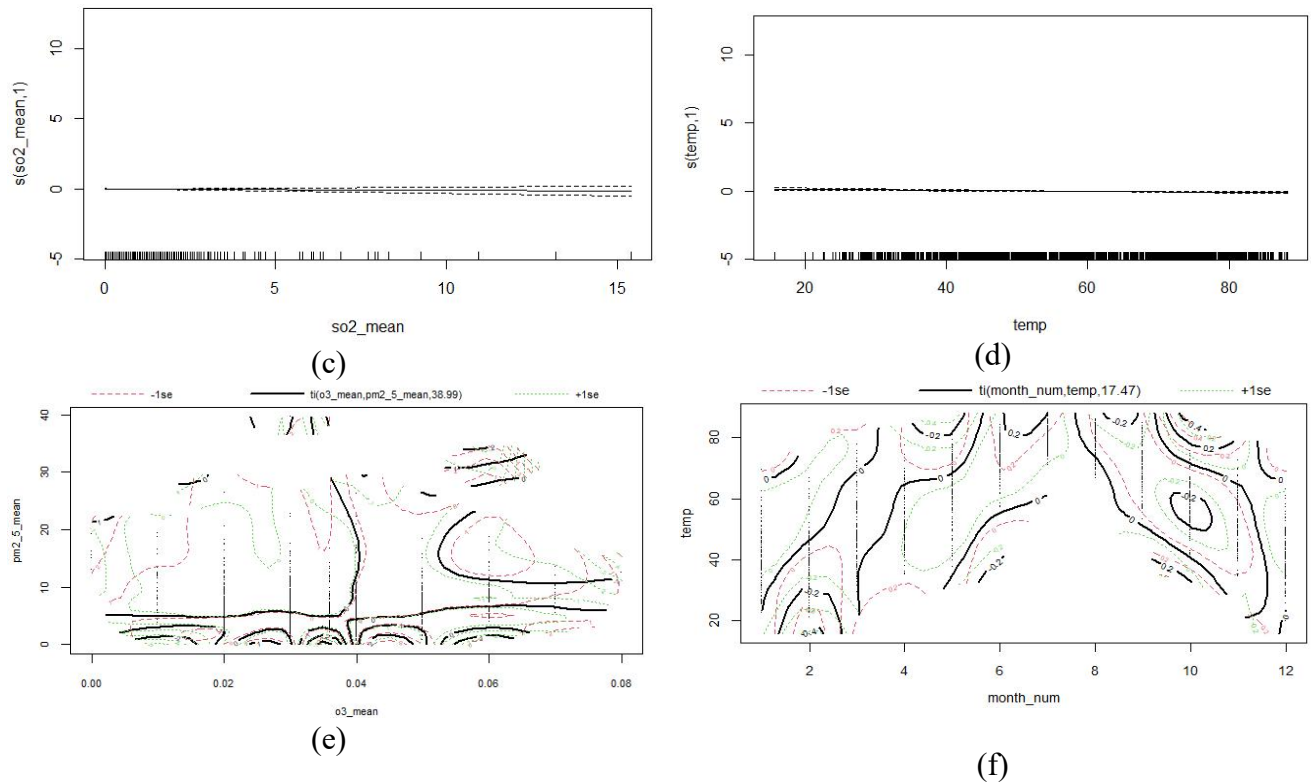


Figure 5: GAM smooth effect plots

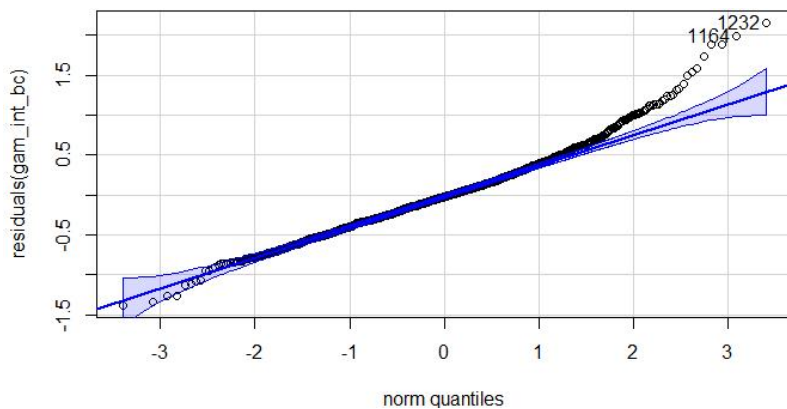


Figure 6 : normal Q-Q plot of the residuals

This is a normal Q-Q plot of the residuals in figure 6. In the middle, the points follow the reference line and stay mostly inside the blue envelope, so residuals are roughly normal for typical AQI values. But in the right tail, several points bend

upward and fall outside the envelope (some are labeled), showing a heavier-than-normal upper tail. The right-tail deviations in the Q-Q plot are driven by a small number of rare extreme AQI days

(very high/rare pollution events, e.g., points 1164/1232), and the model cannot fully fit them-so it leaves large positive residuals. Although such extremes exist in the training year, they are too infrequent and may depend on episodic factors not represented in the predictors, so the model tends to underpredict these events, leaving large positive residuals. Figure 7 demonstrates the Residuals vs fitted and Residuals vs index plots. Residuals are mostly centered around 0 with no strong curve, so the mean structure is captured reasonably well. A bit more spread at mid-high fitted values suggests mild remaining heteroskedasticity/outliers. On the other hand, residuals look roughly random around 0 across the data order, so there's no clear drift/serial pattern. A few isolated large negatives/positives indicate occasional extreme misfits. Overall, these diagnostics suggest the GAM is well-specified and stable, with only minor remaining heteroskedasticity and a few extreme outliers.

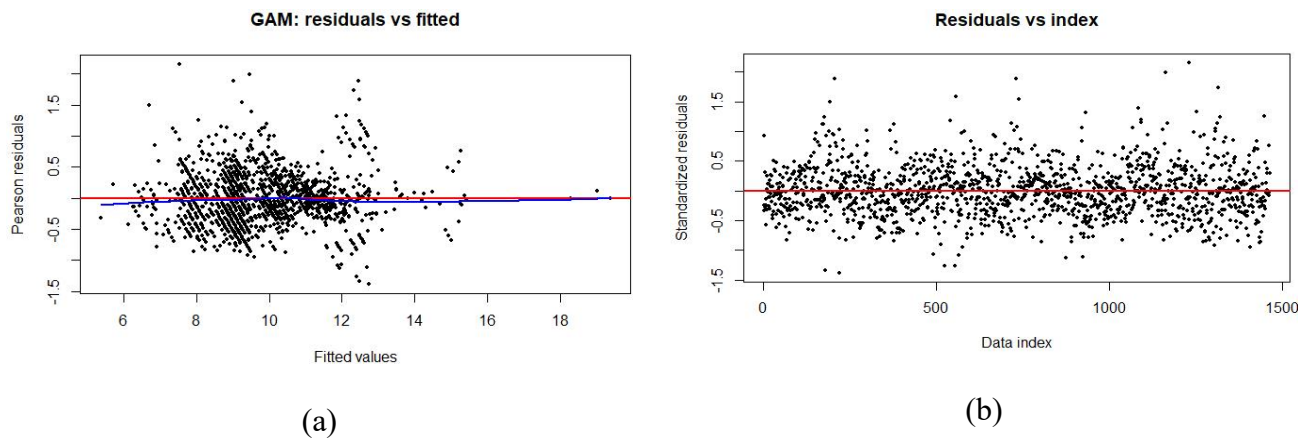


Figure 7: Residual vs Fitted and Index

5 Conclusion

We selected this Box-Cox GAM with cyclic month and tensor interactions between predictors because it was the only model that simultaneously matched the patterns seen in the plots and delivered the best predictive performance. The baseline LM left clear structure in diagnostic plots (curved residual-fitted trends, heteroscedasticity, and heavy right tails in the Q-Q plot), indicating violated linearity/constant-variance assumptions. In contrast, the GAM smooth plots captured the strong nonlinear PM_{2.5} and O₃ effects, the interaction surfaces revealed meaningful O₃×PM_{2.5} synergy and month×temp seasonality, and the Box-Cox transform improved residual normality (remaining departures mainly in rare extreme-AQI tails). Quantitatively, it also achieved the strongest fit (adj. $R^2 = 0.929$; 93.2% deviance explained) and best generalization (RMSE = 4.2775 on 2024), remaining QQ-tail deviations are mainly from rare episodic spikes.